SAMPLE

History of Astronomy: Medieval European Astronomy

Astronomy in the Dark Ages

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Free sample: Introduction & Chapter 1

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Introduction

Between the fall of Rome and the birth of Copernicus lies a thousand-year stretch of European history that most people picture as a scientific desert: an age when the sky was explained by theology, instruments rusted unused, and everyone believed the Earth was flat until an Italian navigator proved otherwise. Every piece of that picture is wrong. The medieval centuries in Europe were not a desert but a slow, stubborn, and ultimately triumphant recovery, and astronomy, the most mathematical and most useful of the sciences, stood at the center of that recovery.

The pages that follow trace astronomy in Europe from the twilight of the Roman Empire to the dawn of the modern age, roughly the fifth century through the fifteenth. It is a story that begins in catastrophe. When the western Empire collapsed, the libraries emptied, the schools closed, and the Greek language of learning was all but forgotten. Astronomy survived the disaster in an unlikely vessel: the prayer schedule of monasteries, which needed to know the hours of the day and the date of Easter, and therefore needed arithmetic of the Sun and Moon even when almost nothing else of ancient science remained. Out of that narrow need grew a tradition, and out of that tradition grew, eventually, the universities, the instruments, the mathematical tables, and the habit of quantitative reasoning that made the scientific revolution possible.

The cast of characters may surprise readers raised on the desert myth. There is the monk of Northumbria who computed the date of Easter so carefully that his arithmetic is still, in a sense, ticking inside the calendars of every Western nation, since our system of counting years from the birth of Christ descends from his tables. There is the pope who had once been a student of mathematics, whose skill with an abacus made his contemporaries whisper that he was a magician. There are the translator-scholars who crossed the Pyrenees to the libraries of Islamic Spain and brought back, one manuscript at a time, the geometry of Euclid and the astronomy of Ptolemy. There is the small-town English schoolmaster whose two-hundred-page textbook on the sphere of the heavens became the most-read science book in Europe for four hundred years. There are the abbot who built a clock of astonishing complexity to display the motions of the heavens, the French master who argued that the Earth might turn without our feeling it, two centuries before Galileo was condemned for saying the same, and the young Austrian who re-observed the ancient sky with such rigor that Copernicus, a generation later, would have real data to stand on.

The story must also be honest about limits. Medieval European astronomy was conservative by design: its highest loyalty was to inherited authority, to Ptolemy, Aristotle, and the Bible, and it rarely questioned the frame of the heavens as a machine of circles. Its observational record was thin compared with what Islamic astronomers had achieved in Baghdad, Cairo, and Maragha, and much of its best work consisted of digesting what other civilizations had discovered. Yet within that conservatism it quietly accomplished things of the first importance: it built the institutions that make organized science possible, it developed mathematical techniques that the later revolution would need, and it kept a tradition of careful calculation alive through centuries when almost nothing else was. The road to Copernicus runs straight through it.

No mathematics beyond ordinary arithmetic is needed for this book, and every technical idea, from the nineteen-year lunar cycle to the gearwork of an astronomical clock, is explained in plain words when it appears. What is assumed is only an appetite for the peculiar: for a world in which the most advanced computing machines in existence were prayer schedules, cathedral clocks, and brass discs engraved with the paths of planets. The people who built those machines were our intellectual ancestors, and their story, long mistold, deserves to be read as it actually happened.

A word on how the journey is organized. The first chapters follow the tradition chronologically, from the salvage of the post-Roman centuries through the Carolingian classroom to the translation flood of the twelfth century. The middle chapters change lenses, examining the age's characteristic practices: how observation was done and recorded, what the instruments and tables actually were, how the church, the economy, and the universities shaped the work. The later chapters assess the inheritance: the fourteenth century's boldest ideas, the puzzles the tradition could not solve, the way modern historians have rebuilt the whole picture, and the era's odd and memorable corners. Along the way, each technical idea, the nineteen-year cycle, the epicycle, the equant, the escapement, is explained at the moment it is needed, so the book can be read straight through without preparation.

Readers who bring a little of the sky with them will find their knowledge honored but not required. The medieval astronomers themselves began as beginners, copying out their tables letter by letter, and one of this book's quiet

arguments is that patient beginnerhood, century after century of it, is itself a scientific method.

Chapter 1: The Discovery of Medieval European Astronomy

The story properly begins with a loss. In the fifth century of our era, the western half of the Roman Empire came apart under invasion, taxation failure, and civil war, and the elaborate infrastructure of ancient learning came apart with it. The great libraries of the Latin West had never rivaled Alexandria's, but they had held the classics of Roman education, and the grammarians and rhetoricians who taught in the Forum had passed on a curriculum that included at least the rudiments of astronomy. As the cities shrank and the roads failed, this apparatus vanished. By the year 600, the number of people in the Latin West who could read Greek, the language of the deepest astronomical works, had fallen nearly to zero, and the last Latin translations of Greek science, shallow and fragmentary, were all that stood between the ancients and oblivion.

Three figures of the sixth and seventh centuries deserve the first credit for saving what could be saved. Boethius, the last great Roman scholar of the West, set himself the task of translating Plato and Aristotle into Latin so that learning would not die with Greek literacy. He did not live to finish the project, executed as he was around 524 on a charge of treason, and of the mathematical works he completed, only the arithmetic and music survived to teach later ages; his intended translation of the astronomy of Ptolemy was never made. Yet his ambition defined the task of the next five hundred years: preserve the ancient curriculum, in Latin, by any means available. Cassiodorus, his contemporary, founded a monastery with an explicit program of copying books, arguing in writing that the copyist serves the preservation of knowledge just as the reader serves its use. Isidore, Bishop of Seville, compiled the Etymologies, a giant encyclopedia that stored, in compressed and sometimes garbled form, whatever the seventh century could still remember about the heavens: the spheres, the planets, the zones of the Earth, the nature of comets and eclipses.

These were salvage operations, not science, and they were carried out in an atmosphere of general catastrophe. But salvage was exactly what the moment required, and the salvage took a decisive turn in the eighth century, when the science of the church calendar demanded real astronomy. The problem was Easter. The Council of Nicaea in 325 had ruled that Easter be celebrated on the Sunday after the first full moon of spring, which sounds simple until one realizes that the date of full moon and the day of the week both had to be known years in advance, over all the parishes of a Europe that had no almanacs and few astronomers. The church therefore commissioned tables, and the construction and defense of those tables became a mathematical discipline in its own right, called the computus, the reckoning.

The master of that discipline was the Venerable Bede, a monk of Wearmouth-Jarrow in Northumbria who never traveled farther from his library than the mouth of the river, and who nevertheless wrote the definitive Western treatise on the calendar, The Reckoning of Time, completed around 725. Bede assembled everything the Latin world still knew about the Sun, the Moon, the tides, and the year, explained the nineteen-year lunar cycle by which the Moon's dates repeat, corrected errors that had crept into older tables, and defended the Roman Easter against rival reckonings with arguments sharp enough to settle the question for good. In the course of the work he also, almost in passing, standardized the counting of years from the incarnation of Christ, the system of Dionysius Exiguus, an obscure Easter-table maker of the sixth century, and gave that system its scholarly credentials. Every date written in this book, and in every modern history, rests on the arithmetic Bede stabilized.

The next step came from politics. Charlemagne, the Frankish king crowned emperor in 800, built a court school at Aachen and staffed it from the learned of Europe, importing Alcuin of York to direct a curriculum in which astronomy, as one of the four mathematical arts, was required of every cleric. His reign's calendar decrees standardized the computus across the empire and put the weight of the state behind correct reckoning. The so-called Carolingian renaissance was bookish, derivative, and modest in scale, but it established the pattern that the whole medieval achievement would follow: institutions, backed by authority, committed to preserving and teaching a fixed body of mathematical learning. Astronomy entered the curriculum in the ninth century as a servant of the calendar, and it never left the curriculum again. When, three centuries later, Europe discovered how much more there was to know, there would already be schools, teachers, and students positioned to learn it.

One further strand of the early centuries deserves rescue from the shadows: the Irish and Northumbrian schools. While the continental survivors gathered fragments, the monks of Ireland, converted without ever being Romanized, built a Latin culture of their own, energetic, textual, and adventurous, whose scholars sailed abroad to found monasteries from Bobbio in Italy to St Gallen in the Alps. Their Easter reckoning, inherited from an older cycle than Rome's, differed by days in some years, and the resulting dispute, argued in Ireland and England and

finally settled at the synod held at Whitby in 664, may look parochial until one sees what it really was: two astronomical theories of the Moon, defended with tables and computations before an audience of kings and abbots, deciding the liturgical calendar of a kingdom. The Easter controversy, in all its phases, was the crucible in which Western computational astronomy was forged, and the Northumbrian school that gave the world Bede was its final winner.